

ARTIFICIAL INTELLIGENCE & GENERATIVE AI

AI Across the Business

Where AI creates real value in marketing, finance, HR and operations – and what AI agents change about how work gets done.

MARKETING

FINANCE

HR

OPERATIONS

AI AGENTS

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What we will cover

Function by function, then the technology that ties it together.

o1

Where AI pays off

The four questions that spot a good use case

o2

AI in Marketing

Targeting, content and campaign decisions

o3

AI in Finance

Forecasting, fraud, reporting and risk

o4

AI in HR

Hiring, onboarding, engagement and retention

o5

AI in Operations

Demand, supply chain, quality and maintenance

o6

What is an AI agent?

The step beyond a chat assistant

o7

Types of agents

Six kinds, from simple rules to teams of agents

o8

How an agent works

The six-step loop behind every agent

SECTION 01

Where AI Actually Pays Off

Not every problem needs AI. Some are perfect for it.

A good AI use case usually has all four

Fail any one of these and a spreadsheet or a clear process is the better answer.

It repeats often

Hundreds or thousands of times a month. Rare tasks are not worth automating.

There is data

History exists in a system already – not only in people's heads.

Rules are fuzzy

Judgement is involved, so a simple 'if this then that' rule does not work.

Mistakes are survivable

A wrong answer can be caught and corrected before it does damage.

AI creates value in four ways

Every business case eventually lands in one of these buckets.

- Save time – the work gets done faster
- Cut cost – fewer hours for the same output
- Grow revenue – better targeting and pricing
- Reduce risk – catch problems earlier

■ Save time

Draft, summarise and sort work that used to be manual.

■ Cut cost

Handle routine volume without adding headcount.

■ Grow revenue

Right offer, right customer, right moment.

■ Reduce risk

Spot fraud, defects and churn before they land.

SECTION 02

AI in Marketing

The function where adoption started, and moved fastest.

Six things marketing teams use AI for

■ Customer segments

Groups customers by real behaviour instead of broad age and location buckets.

■ Content at scale

First drafts of ads, emails, posts and product descriptions in every variant.

■ Personalisation

Different message and offer for each segment, automatically.

■ Send-time and channel

Predicts when and where each customer is most likely to respond.

■ Lead scoring

Ranks enquiries so sales calls the most promising ones first.

■ Campaign analysis

Explains which channel actually drove the result, not just the last click.

What good looks like

Same team, more output, better targeting.

Days to hours

Campaign production

Draft copy and creative variants are ready the same day the brief lands.

I-to-I

Personalisation

Every customer can get a relevant message, not one mass email.

Fewer wasted calls

Sales efficiency

Lead scoring puts the best prospects at the top of the list.

The risk: everyone's marketing starts to sound the same. Brand voice and human editing matter more than ever.

SECTION 03

AI in Finance

Where accuracy, speed and audit trails all matter.

Six high-value finance applications

Mostly predictive AI, with Gen AI doing the writing.

■ Forecasting

Predicts revenue, cash flow and cost using history, seasonality and market signals.

■ Fraud detection

Flags transactions that break the customer's normal pattern, in real time.

■ Credit risk

Scores applicants on far more signals than a manual review could handle.

■ Invoice processing

Reads invoices, matches them to purchase orders and routes exceptions.

■ Reporting

Drafts the narrative around the numbers so analysts review instead of write.

■ Audit support

Scans every transaction for anomalies rather than checking a sample.

Fraud detection: before and after

RULES ONLY

Fixed thresholds written by people

- Blocks anything over a set amount
- Fraudsters learn the limits and stay under them
- Large numbers of false alarms
- Every new pattern needs a new rule
- Genuine customers get blocked

WITH AI

Learns each customer's normal behaviour

- Compares to that customer's own history
- Adapts as new fraud patterns appear
- Far fewer false alarms
- Learns continuously from outcomes
- Still needs a human for the edge cases

IN SHORT AI does not replace the rules. It handles what the rules cannot describe.

SECTION 04

AI in Human Resources

High value, and the highest need for care.

Six ways HR teams use AI

■ Job descriptions

Consistent, inclusive postings drafted in minutes.

■ CV screening

Shortlists against the skills that actually matter for the role.

■ Interview support

Suggests questions and structures notes so candidates are compared fairly.

■ Onboarding help

An assistant that answers policy questions on day one, at any hour.

■ Engagement insight

Finds the themes in thousands of open survey comments.

■ Attrition risk

Flags teams where people are likely to leave, so managers can act early.

HR is the highest-risk place to use AI badly

Use AI to widen the funnel and save admin time, not to pick people.

- **Bias is inherited**

A model trained on past hiring will copy past hiring bias.

- **Decisions must be explainable**

A rejected candidate may be entitled to know the reason.

- **Data is deeply personal**

Salary, performance and health data need strict access control.

- **Regulators are watching**

Hiring AI is treated as high risk in most new AI laws.

- **Screen, do not decide**

AI can shortlist. A person must make the final call.

- **Tell people it is in use**

Transparency avoids most of the trust problems later.

SECTION 05

AI in Operations

The quietest wins, and often the biggest.

Six operational applications

■ Demand forecasting

Predicts what will sell where, so stock sits in the right place.

■ Inventory planning

Balances holding cost against the cost of running out.

■ Route optimisation

Plans delivery routes around traffic, weight and time windows.

■ Predictive maintenance

Flags a machine that is about to fail, before it stops the line.

■ Quality inspection

Cameras spot defects faster and more consistently than tired eyes.

■ Service automation

Handles routine customer questions and escalates the rest.

One view of all four functions

FUNCTION	TYPICAL USE	MAIN BENEFIT	MAIN RISK
Marketing	Segments, content, personalisation	Speed and relevance	Generic, off-brand output
Finance	Forecasting, fraud, reporting	Accuracy and early warning	Over-trusting a model
HR	Screening, onboarding, retention	Less admin, wider funnel	Bias and legal exposure
Operations	Demand, routes, maintenance	Lower cost, less downtime	Bad data breaks the forecast

Different functions, same pattern: AI drafts and predicts, people decide and own it.

SECTION 06

AI Agents

From answering questions to getting things done.

An assistant answers. An agent acts.

Same underlying model. Very different amount of autonomy.

CHAT ASSISTANT

You do the work, it helps you think

- Waits for each instruction
- Produces text and nothing else
- Forgets once the chat closes
- Cannot touch other systems
- You carry out every step yourself

AI AGENT

You set the goal, it works towards it

- Given an objective, not a script
- Plans the steps by itself
- Uses tools: email, search, databases
- Checks its own progress and retries
- Reports back when the goal is met

IN SHORT An assistant helps you work. An agent does the work and reports back.

Six types, in order of independence

More independence means more value, and more supervision needed.

1

■ Simple reflex

Reacts to what is happening right now. No memory. A thermostat.

2

■ Model-based

Keeps an internal picture of the situation, so it can handle what it cannot directly see.

3

■ Goal-based

Given an outcome, it plans a route to reach it.

4

■ Utility-based

Weighs trade-offs and picks the best option, not just a valid one.

5

■ Learning

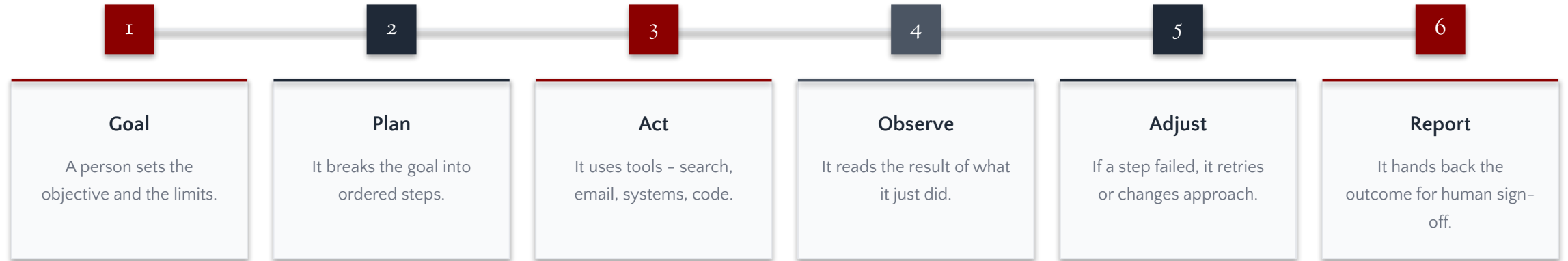
Improves from feedback and results over time.

6

■ Multi-agent

Several specialist agents working together with a coordinator.

The six-step loop inside every agent



The loop is what makes it an agent. Plan, act, check, adjust – repeat until done.

A support agent, end to end

One goal, six steps, one human check at the end.

- Goal: resolve this refund request
- Reads the ticket and finds the order
- Checks the refund policy and the delivery date
- Drafts the reply and prepares the refund
- Escalates to a person if the amount is large
- Logs everything for the audit trail

■ What people gain

Routine tickets close in minutes, at any hour.

■ What people keep

The judgement calls, exceptions and the relationship.

■ What you must add

Clear limits, an audit log and an easy way to escalate.

Four guardrails every agent needs

Autonomy without guardrails is how pilots turn into incidents.

■ Clear limits

- Define what it may never do alone
- Cap the value it can approve
- Restrict which systems it can reach

■ Human checkpoints

- Approval before anything irreversible
- A named owner for every agent
- An obvious way to stop it

■ Full logging

- Record every action and decision
- Keep it reviewable and auditable
- Know why it did what it did

■ Start small

- One narrow task, low risk, first
- Measure quality before widening scope
- Expand only when it has earned trust

Key takeaways

Unit 3 in six lines.

1

Good use cases repeat, have data, need judgement and tolerate error

Check all four before you invest.

2

Marketing gains speed and relevance

Segments, content and personalisation at scale.

3

Finance gains accuracy and early warning

Forecasting, fraud, invoices and reporting.

4

HR gains time but carries the most risk

Screen and support. Never let AI make the final call.

5

Operations gains cost and uptime

Demand, routes, maintenance and quality.

6

Agents act, so they need guardrails

Limits, human checkpoints, logging, and a small start.

CONCLUSION

Thank you

Next up — Unit 4: privacy, security, regulation and what happens when AI starts influencing real decisions.

QUESTION 1

Which function in your organisation would gain the most, fastest?

QUESTION 2

Which decision would you never hand to an agent?

QUESTION 3

What data would you need before any of this could work?